

Arunabha Majumder

Prototype Engineer | Wearable Robot Hardware, Test Rigs, 3D Printing and Assembly

+45 71 62 24 00 | Aalborg, Denmark (open to relocating to London) | somrkmv1997@gmail.com |

97arunabhasit027@gmail.com

LinkedIn: linkedin.com/in/arunabha-majumder-681264107 | Portfolio: arunabha22.github.io/arunabha-majumder |

Project: www.viexo.aau.dk

SUMMARY

I build robot hardware that people wear, and the rigs to test it. For three years I have turned my own CAD files into working shoulder exoskeleton units: printing the parts, putting in the motor, encoder and sensors, fixing what failed, and running it with real users in trials. I like a messy bench that produces a working part by Friday. SolidWorks (CSWA), 3D printing, Arduino, MATLAB and Python.

KEY RESULTS

- **12** human trial participants run on hardware I built and kept working
- **>25%** less energy from the spring and motor exoskeleton I designed and built (Best Paper, 2026)
- **<10%** force error on a test setup I built, checked against a force sensor

EXPERIENCE

Research Engineer, Wearable Robot Prototypes (PhD project VIEXO) | 2023 - Present

Aalborg University, Aalborg, Denmark

- Kept a human-worn shoulder exoskeleton running through trials with 12 participants, 3 assist modes and 2 loads by building the units from my own CAD files and BOMs, and repairing and refitting them between sessions.
- Went through several versions of each part in weeks rather than months by 3D printing them in house, fitting them on the rig the same day and sending what I learned straight back into the CAD.
- Cut energy use by more than 25% (Best Paper Award, IFToMM ISRM 2026) with a spring plus 6 Nm motor actuator that I designed in SolidWorks, then built and assembled myself.
- Got force readings within 10% RMS of a reference sensor by building a test setup around a 5-bar parallel robot with variable-stiffness actuators and calibrating it against a force sensor.
- Wired motors, encoders, force sensors and DAQ into test rigs and an Arduino-based setup for functional, calibration and durability checks, and helped run lab demos. Supervised Bachelor's students (2023 - 2025).

MTech Researcher, Pneumatic Actuators (Master's thesis) | 2020 - 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Made pneumatic artificial muscles by hand from raw materials and built the test bench around them (valves, pressure sensors, DAQ), then used it to tune a controller that brought position error down to 0.3-0.78%.

Bachelor's Capstone and Student Builds | 2017 - 2019

Siddaganga Institute of Technology, Tumakuru, India

- Designed, machined, assembled and user-tested an upper-limb rehabilitation exoskeleton. Best Major Project Award and an INR 100,000 TATA Technologies grant.
- Built a Go-Kart from scratch (2017) and a human-electric hybrid vehicle for Efficycle (2018) in the college workshop.

SKILLS

- **Workshop:** 3D printing (Bambu Lab, Ultimaker Cura), hand tools, machining and assembly (Bachelor's), test rigs and jigs
- **CAD:** SolidWorks (CSWA): reading and making parts, fixtures, assemblies and BOMs
- **Electronics and MCUs:** Arduino, motors, encoders, force/torque and pressure sensors, DAQ, pneumatic valves
- **Software:** MATLAB/Simulink, Python, LabVIEW (basic) for test scripts and data

- **Languages:** English (C1, fluent), Bengali (native), Hindi (conversational)

EDUCATION

PhD, Mechatronics and Robotics (in progress) | 2023 - Present

Aalborg University, Denmark

MTech, Mechatronics | 2020 - 2022

AcSIR - CSIR-CMERI, Durgapur, India

BTech, Mechanical Engineering | 2015 - 2019

Siddaganga Institute of Technology (VTU), India

AWARDS AND PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Majumder, A., et al. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." ISRM 2026, Springer. Also IEEE CONECCT 2022 (pneumatic muscle control).

REFERENCES

Available upon request.